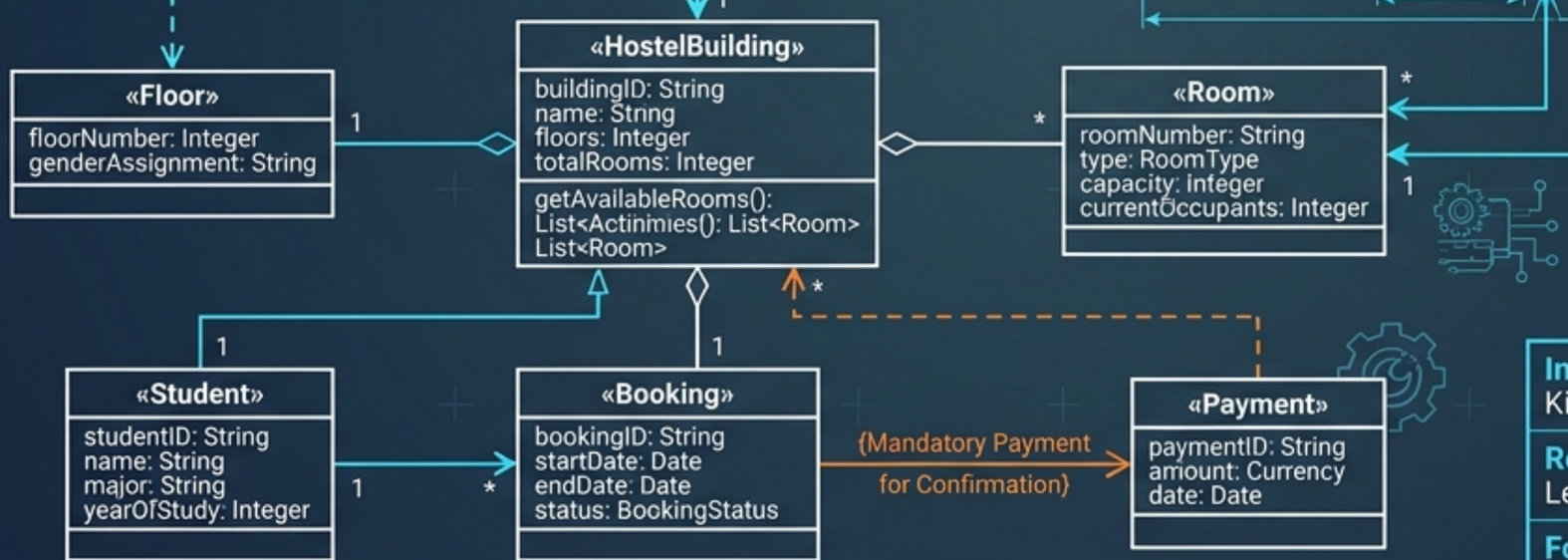
Campus Hostel Booking System: OOAD Blueprint

SOEN 201 Object-Oriented Analysis & Design | Complete Assignment Schematic



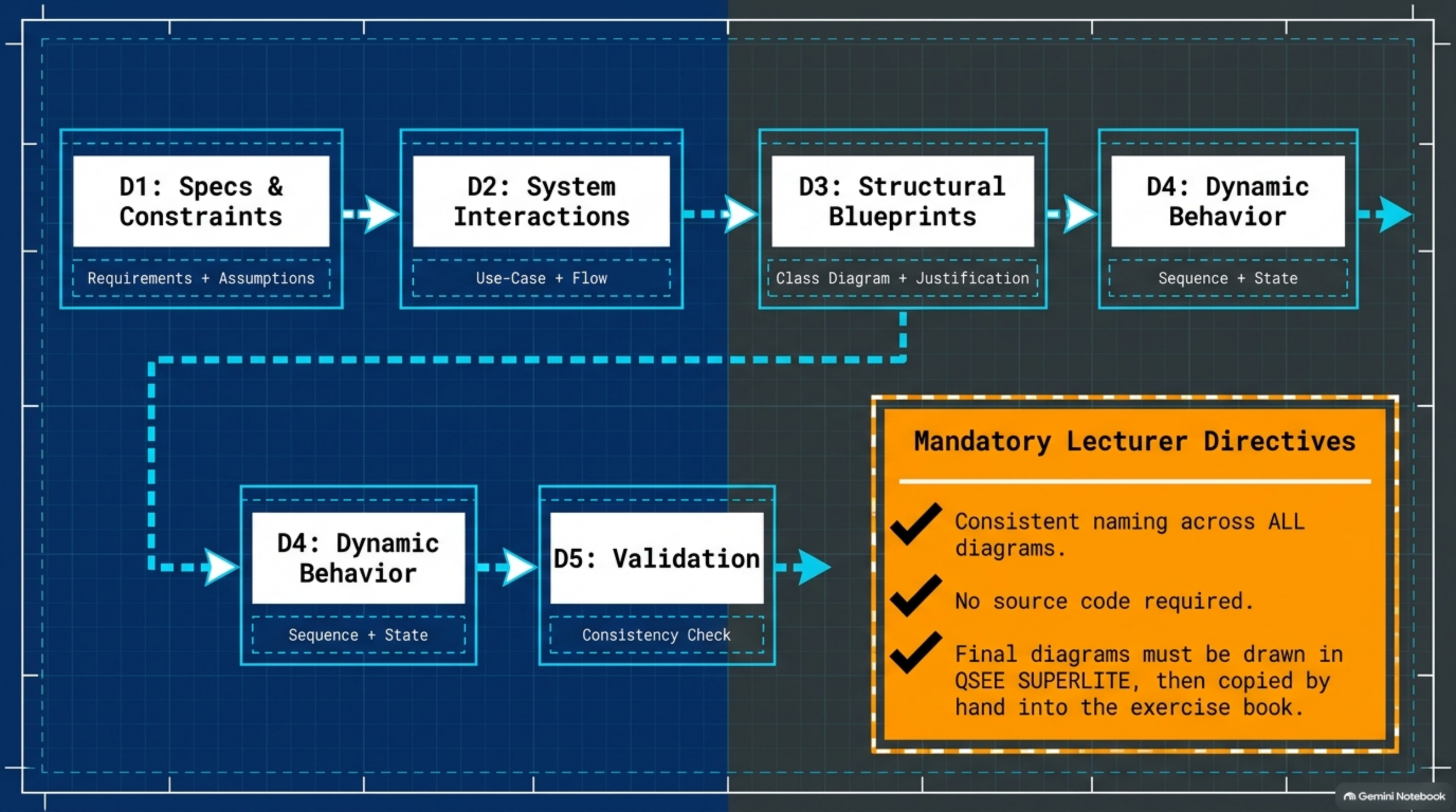


Institution:
Kisii University (School of Computing & IT)

Reference Specs:
Lecturer OOSAD Slides 1–233 + UML Building Blocks

Format:
QSEE SUPERLITE -> Manual Transcription

Gemini Notebook



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Translating the Scenario to UML Structure

Diagram Taxonomy Matrix	
Structural Blueprints (Class Diagram) 	**Behavioral Dynamics (Use Case / Sequence / State)** 
- **Halls & Rooms:** Single/double/four-person, gendered vs. mixed, caretakers. - **Entities:** Students, Applications, Allocations, Waiting Lists.	- **Application Phase:** Opening/closing dates, requests (hall, type, roommate). - **Allocation Phase:** Priority ranking, 7-day deadlines, capacity exceptions. - **Post-Allocation:** Transfers, early leaves, room repairs.

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System Specification Sheet: Housing Allocation Requirements



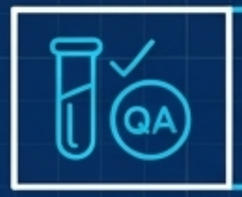
[R1] Limit: The system **shall** allow exactly one application per semester before closing date.



[R2] Preferences: The system **shall** record preferred hall, room type, and optional roommate.



[R3] Validity: The system **shall** allocate only when free, not under repair, and respects gender policy.



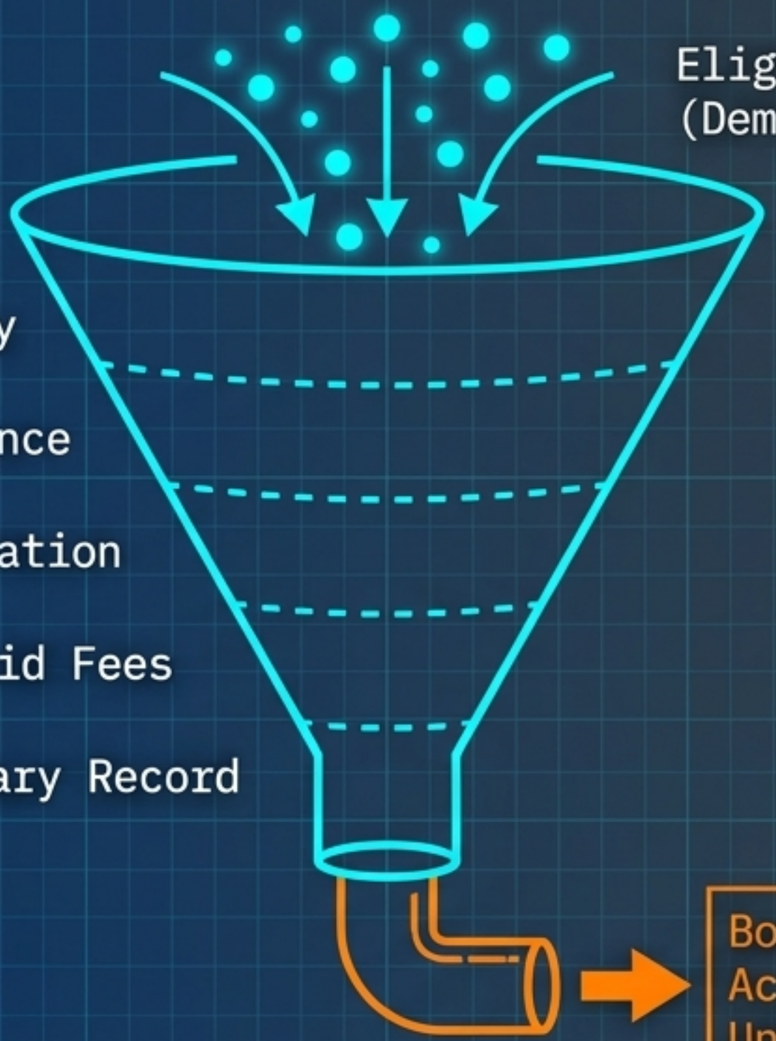
[R4] Deadlines: The system **shall** issue an offer with a deadline; unconfirmed offers **shall** lapse to the to the available pool.



[R5] Waitlist: The system **shall** waitlist students when demand exceeds capacity and notify when available.

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Assumption A1: The Priority Allocation Rule



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Assumption A2: 7 Calendar Days

The system automatically lapses unconfirmed offers after this period.



Assumption A3: One Active Application

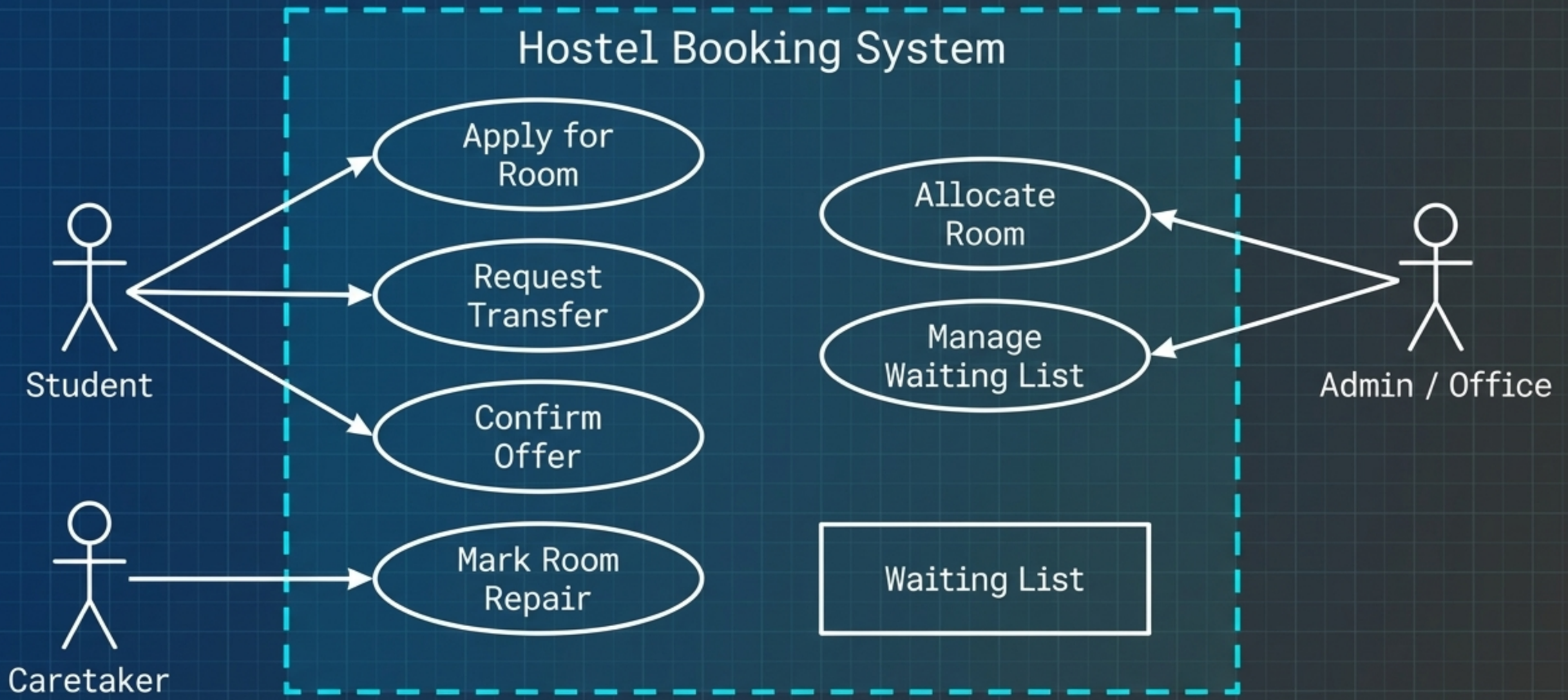
Students cannot apply again until the previous application is withdrawn, allocated, or rejected.

Lecturer Note: Always explicitly label assumptions to avoid losing marks. These define system boundaries not explicitly given in the brief.

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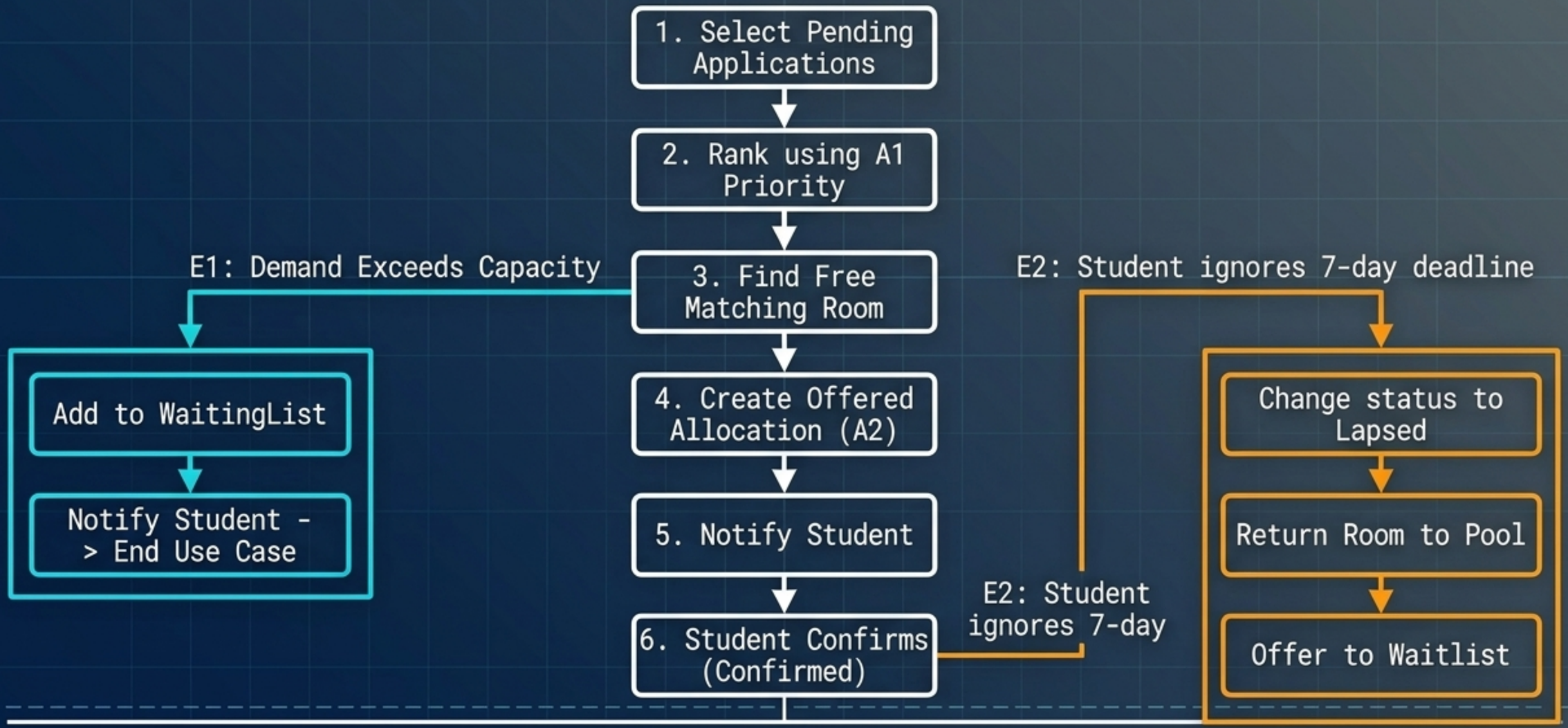
Hostel Booking System

Rule: Actors never cross the boundary. Use correct standard UML hollow ellipses for use cases.



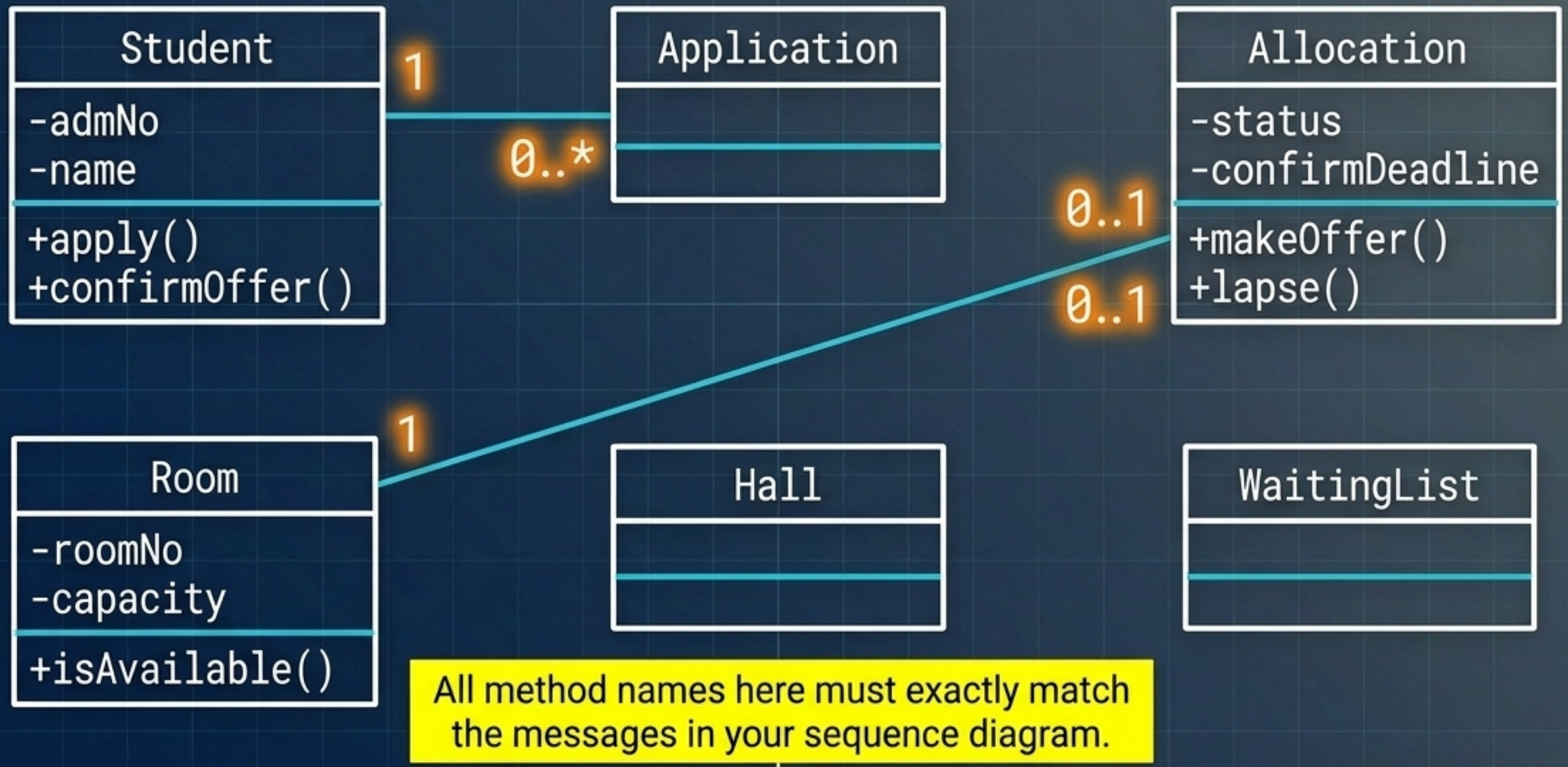
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Allocate Room: Main Success Scenario & Exception Flows



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Simplified cass with: High-contrast UML Class Diagram



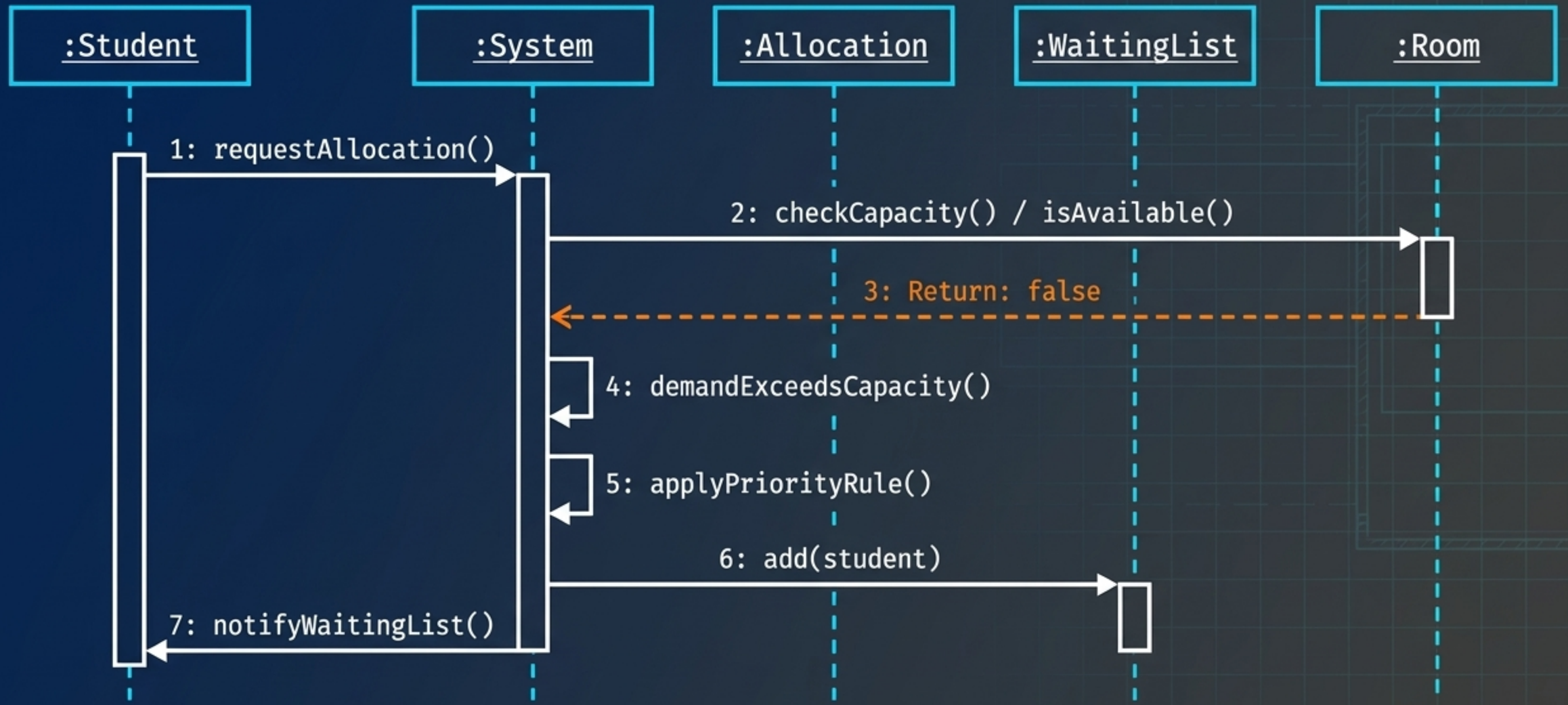
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The Allocation Dilemma: Attribute vs. Class

	As an Attribute (X)	As a Class (✓)
Identity	Just a value.	Has its own unique identifier and dates.
Lifecycle/State	Static.	Transitions: **Offered -> Confirmed -> Lapsed**.
Associations	Cannot link to objects.	Links **Room**, **Application**, and **WaitingList**.
Owned Operations	Cannot act.	Owns **makeOffer()**, **lapse()**, **release()**.

Verdict: Allocation must be modeled as a first-class structural element (Class).

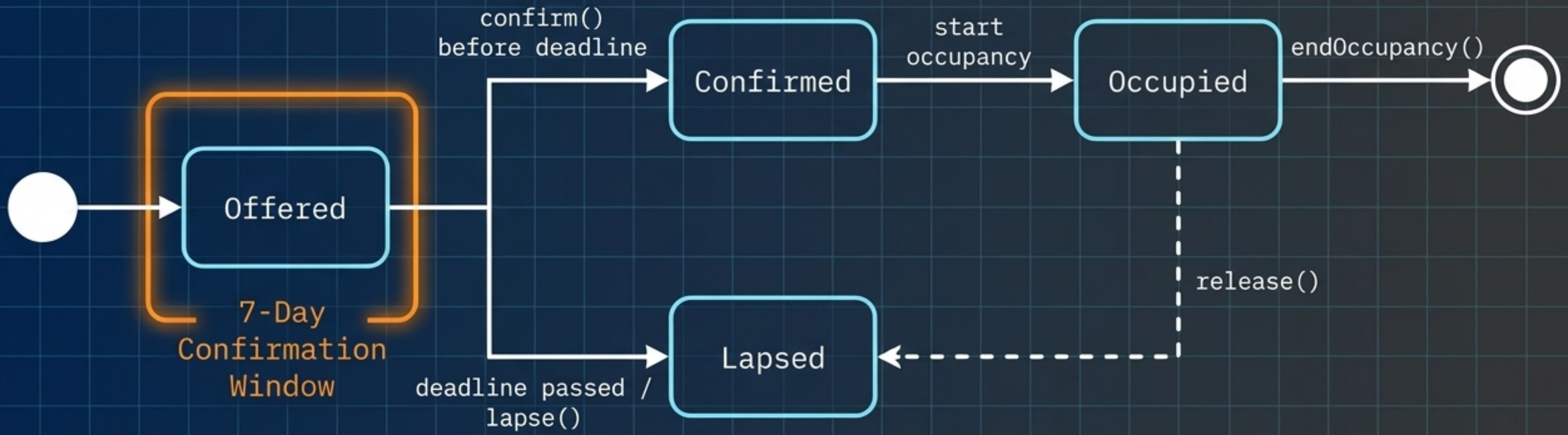
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Notice: No Allocation object is created in this sequence because the room check returned false.

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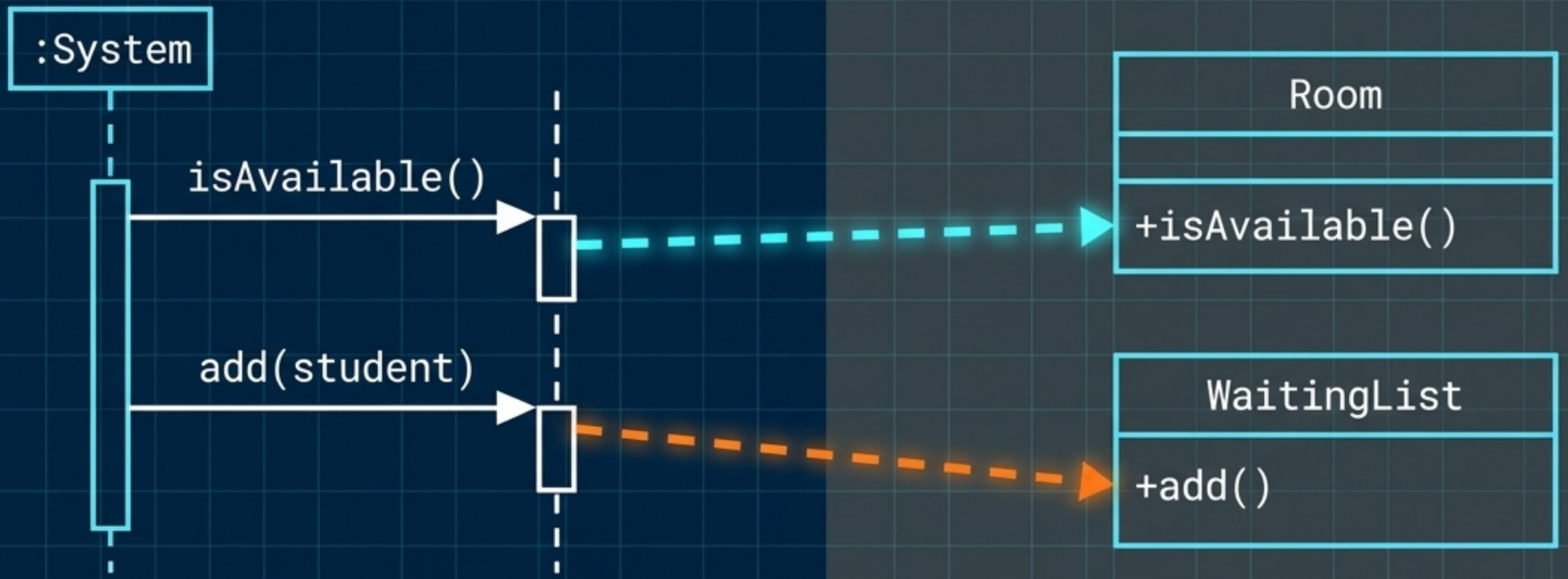
Allocation Lifecycle: UML State Machine Diagram



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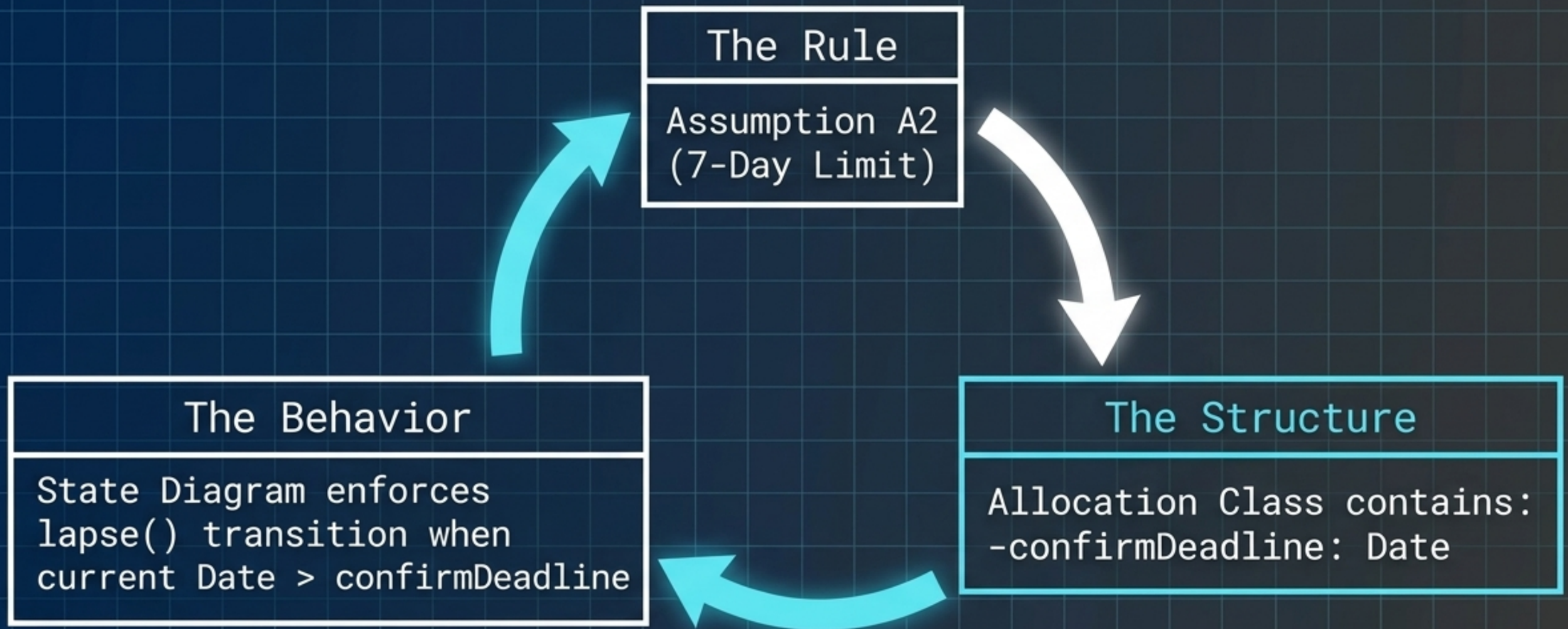
Behavioral Interactions

Structural Operations



Consistency Check 1: Every message fired in the sequence timeline perfectly matches a declared operation in the system's structural blueprint.

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Consistency Check 2: The deadline isn't just a rule—it is structurally embedded as an attribute and behaviorally executed as an automated state transition.

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Architect’s Sign-Off Sheet

- [ ] 5 testable “shall” requirements
- [ ] 3 labeled Assumptions **(incl. Priority Funnel)**
- [ ] Use-Case Diagram + Detailed Allocate Room
- [ ] Class Diagram + “Allocation is a Class” proof
- [ ] Sequence Diagram **(Waitlist Exception)**
- [ ] State Diagram **(Offer Lifecycle)**
- [ ] Consistency linking paragraph
- [ ] **ALL names exactly identical across all diagrams**

Proceed to manual drafting. Copy finalized QSEE SUPERLITE schematics by hand into the exercise book. Include all group members’ names & admission numbers.

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